EXPERIMENT 5: FOURIER SERIES APPROXIMATION OF PERIODIC SIGNALS

Input Coding

```
clc;
clear;
close all;

% Time vector
t = linspace(-2*pi, 2*pi, 4000);

% Original bipolar square wave
x = sign(sin(t));

% Highest harmonics to be used
N_values = [3 7 15];

% Create figure
figure;

for i = 1:length(N_values)

   N = N_values(i);

   % Initialize Fourier reconstruction
   xN = zeros(size(t));

   % Fourier Series of bipolar square wave
   % x(t) = (4/pi)[sin(t) + sin(3t)/3 + sin(5t)/5 + ...]

   for k = 1:2:N
      xN = xN + (4/pi)*(1/k)*sin(k*t);
   end

   % Plot reconstruction
   subplot(3,1,i);
   plot(t, x, 'k--', 'LineWidth', 1);
   hold on;
   plot(t, xN, 'LineWidth', 1.5);
   grid on;

   xlabel('Time (t)');
   ylabel('Amplitude');
   title(['Bipolar Square Wave - Highest Harmonic = ', num2str(N)]);
   legend('Original', 'Fourier Reconstruction');

end
```

```
sgtitle('Fourier Series Approximation of Bipolar Square Wave');

%% Zoom near discontinuity
figure;
for i = 1:length(N_values)

   N = N_values(i);

   xN = zeros(size(t));

   for k = 1:2:N
      xN = xN + (4/pi)*(1/k)*sin(k*t);
   end

   subplot(3,1,i);
   plot(t, xN, 'LineWidth', 1.5);
   grid on;

   xlim([-0.5 0.5]);
   ylim([-1.5 1.5]);

   xlabel('Time (t)');
   ylabel('Amplitude');
   title(['Near Discontinuity, Highest Harmonic = ', num2str(N)]);

end

sgtitle('Gibbs Phenomenon Near Discontinuity');

%% Fourier Coefficient Decay
figure;
hold on;

for N = N_values

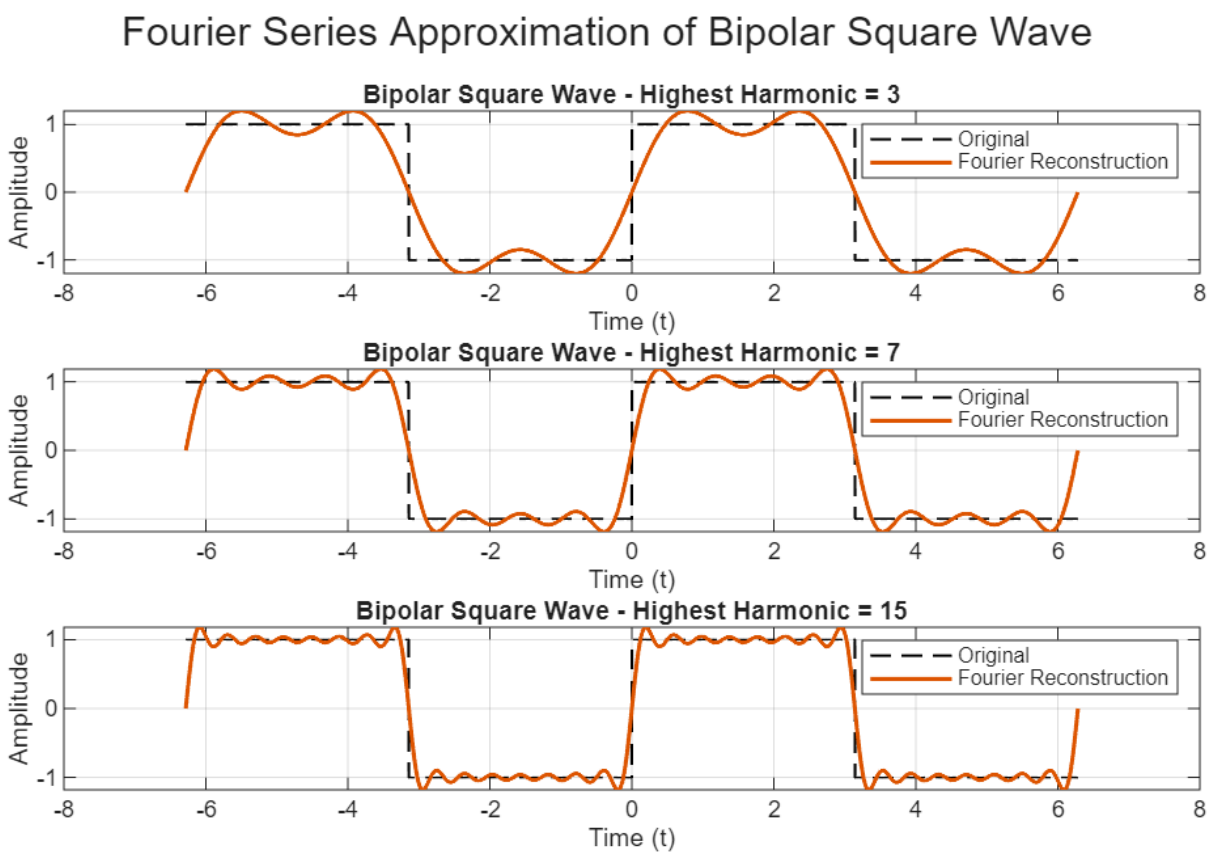
   k = 1:2:N;
   coefficients = 4./(pi*k);

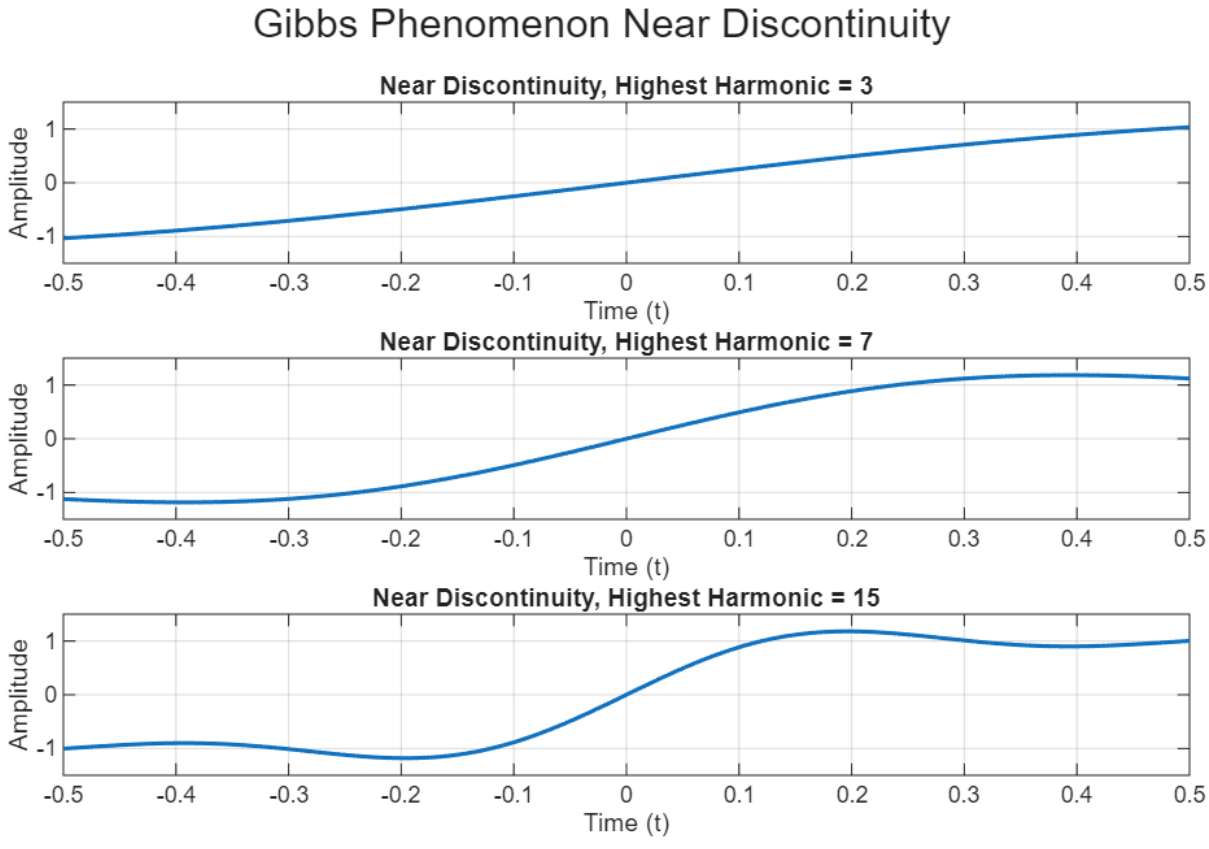
   stem(k, coefficients, 'filled');

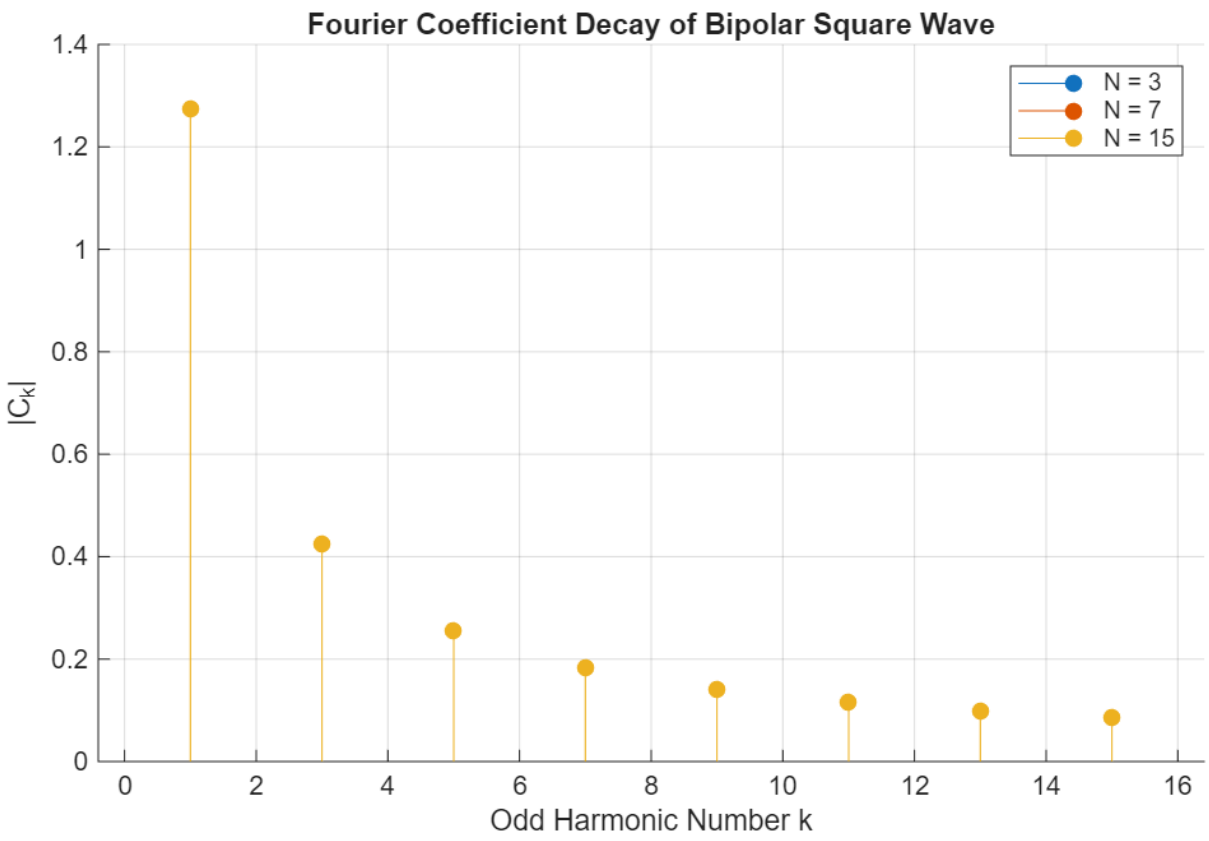
end

grid on;
xlabel('Odd Harmonic Number k');
ylabel('|C_k|');
title('Fourier Coefficient Decay of Bipolar Square Wave');
legend('N = 3', 'N = 7', 'N = 15');
```

OUTPUT


Fourier Series Approximation of Bipolar Square Wave
Bipolar Square Wave - Highest Harmonic = 3
Bipolar Square Wave - Highest Harmonic = 7
Bipolar Square Wave - Highest Harmonic = 15
Original
Fourier Reconstruction
Amplitude
Time (t)


Gibbs Phenomenon Near Discontinuity
Near Discontinuity, Highest Harmonic = 3
Near Discontinuity, Highest Harmonic = 7
Near Discontinuity, Highest Harmonic = 15
Amplitude
Time (t)


Fourier Coefficient Decay of Bipolar Square Wave
N = 3
N = 7
N = 15
1.4
1.2
1
0.8
0.6
0.4
0.2
0
|Ck|
0
2
4
6
8
10
12
14
16
Odd Harmonic Number k